BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND ENGINEERING

Discriminative Prompt Rerouting and Query Prototype Refinement on Calibrated Prototype in Frozen Few Shot Class Incremental Learning

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Md Akib Haider**, **Fateen Noor Rafee**, and **Hasib Altaf** under the supervision of **Reaz Hassan Joarder**, Junior Lecturer, Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

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Dedicated to our families and teachers, whose patience and encouragement made this work possible

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List of Abbreviations

BiMC	Bi-level Modality Calibration
CLIP	Contrastive Language-Image Pre-training
CUB	Caltech-UCSD Birds
DINO	Self-Distillation with No Labels
DPR	Discriminative Prompt Rerouting
FSCIL	Few-Shot Class-Incremental Learning
LLM	Large Language Model
PD	Performance Degradation
QPR	Query Prototype Rectification
TRiC	Tri-Level Calibration
t-SNE	t-distributed Stochastic Neighbor Embedding
ViT	Vision Transformer
VLM	Vision-Language Model

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Abstract

Few-shot class-incremental learning (FSCIL) requires a system to learn a long stream of new classes from only a handful of labeled images per class while retaining accuracy on a much larger set of base classes learned earlier, and it must do so without revisiting old data. Recent training-free methods sidestep catastrophic forgetting by keeping every encoder frozen and assembling the final classifier from pre-computed prototypes. This convenience exposes three problems that a calibrated pipeline built on a single vision-language backbone cannot solve: a self-supervised visual encoder such as DINOv2 lives in an embedding space that is geometrically incompatible with that of CLIP, every novel-class prototype is estimated from as few as five images, and the semantic prototypes of closely related classes are built independently of one another and can end up nearly indistinguishable. This thesis addresses all three. First, we present Tri-Level Calibration (TRiC), a training-free framework that separates classifier assembly into a semantic calibration, an inter-model calibration that fuses CLIP and DINOv2 evidence on a common class-score axis, and an inter-modal calibration of semantic and visual evidence. Second, we present Query Prototype Rectification (QPR), a label-free module that refines only the noisy novel-class DINOv2 prototypes with unlabeled queries admitted by a precision-favoring posterior gate. Third, as the principal contribution of this thesis, we propose Discriminative Prompt Rerouting (DPR), a label-only module that builds a directional matrix of pairwise contrastive prompts, repeatedly selects the most confusable class pair, and reroutes each semantic prototype toward its discriminative prompt embedding only when two guards confirm that the pair becomes less confusable and that no other class becomes more confusable. Across *mini*ImageNet, CIFAR-100, and CUB-200, the tri-level backbone improves final-session accuracy over the strongest training-free baseline by 1.32, 9.83, and 12.53 points, and QPR adds a further 1.01, 2.43, and 0.88 points. On CIFAR-100, DPR raises the average session accuracy of the fused backbone from 85.73% to 86.10%, lowers performance degradation from 6.68 to 6.21 points, and improves every incremental session by between 0.24 and 0.54 points, while a prototype-level visualization shows that the rerouting also lowers the rate at which novel queries are absorbed by base classes.

Chapter 1

Introduction

1.1 Background and Motivation

Standard image classifiers are trained once on a fixed set of classes, but real deployments keep meeting new categories, often with only a handful of labeled examples each, such as rare wildlife species or newly registered products. Few-shot class-incremental learning (FSCIL) formalizes this setting [23]. A model is first trained on a base session with many labeled examples per class and then receives a sequence of incremental sessions, each adding a few new classes with only K labeled images per class (typically $K = 5$). After every session it must classify queries from all classes seen so far, without revisiting earlier training data.

Two difficulties follow. Adapting parameters to new, scarce data overwrites what was learned for old classes, a phenomenon known as catastrophic forgetting [8], [18]. Each new class is also learned from a small and possibly unrepresentative sample, so every session is a few-shot problem as well [4], [22]. Most FSCIL methods control forgetting through carefully regularized gradient updates [20], [28], [32], which still require training at every session. Training-free methods instead keep every weight frozen and build each session's classifier from class prototypes [11], [27], so forgetting of the base classes is impossible by construction. Recent training-free methods rely on vision-language models such as CLIP [19], which can build a semantic prototype from a class name alone, and BiMC [5] combines such semantic prototypes with CLIP visual prototypes in a bi-level calibration.

This bi-level design leaves three problems. First, a self-supervised encoder such as DINOv2 [17] provides stronger visual features than CLIP, but its embedding space is geometrically incompatible with CLIP's, so the two cannot simply be averaged. Sec-

ond, every novel-class prototype is estimated from as few as five images and remains noisy after calibration. Third, each semantic prototype is built from its own class name and descriptions, independently of every other class, so close relatives such as two sighthound breeds or two gull species can receive nearly identical semantic prototypes.

This thesis addresses the three problems in turn. Tri-Level Calibration (TRiC) fuses CLIP and DINOv2 evidence on a common class-score axis. Query Prototype Rectification (QPR) refines the noisy novel prototypes with confidently predicted unlabeled queries. Discriminative Prompt Rerouting (DPR), the central contribution of this thesis, reroutes the semantic prototypes of the most confusable class pairs apart using prompts written to contrast each pair, and it needs no images at all.

1.2 Problem Statement

Given a frozen, pre-trained vision(-language) backbone and a stream of FSCIL sessions in which each new session contributes only a handful of labeled images per new class, this thesis addresses the following problem: *how should evidence from multiple, geometrically incompatible frozen encoders (a vision-language encoder and a self-supervised visual encoder), together with class-level semantic evidence (prompts and descriptions) and session-level unlabeled evidence (the query batch), be combined into a single classifier so that (i) base-class accuracy degrades as little as possible across sessions, (ii) novel-class accuracy is as high as possible despite each novel class being supported by only a handful of images, and (iii) no gradient-based training, and no additional human annotation, is required at any point after the base session?*

Concretely, this thesis pursues four specific objectives:

1. Formalize frozen FSCIL as a tri-level calibration problem, separating semantic calibration, inter-model visual calibration, and inter-modal calibration into three explicit, independently controllable steps, and empirically justify why this separation is required once a second visual encoder is introduced alongside a vision-language encoder.

2. Design a label-free rectification mechanism that consumes only the unlabeled query batch already available in a deployed FSCIL system, and uses it to reduce the residual noise in five-shot novel-class visual prototypes without risking base-class confusion.

3. Design a label-only rerouting mechanism that consumes only class names and

natural-language prompts, with no images at all, to reduce semantic confusion between pairs of classes whose prototypes are highly similar, and measure its effect on session-wise accuracy and performance degradation.

4. Empirically characterize the resulting system across three FSCIL benchmarks of increasing granularity—*mini*ImageNet, CIFAR-100, and the fine-grained CUB-200 bird dataset—at the level of session-wise accuracy, Full/Base/Novel accuracy, and gate diagnostics.

1.3 Research Challenges

Several challenges specific to this problem setting motivate the design choices made throughout this thesis.

Cross-encoder geometric incompatibility. A CLIP visual prototype and a DINOv2 visual prototype for the same class are not directly comparable vectors: they differ in norm scaling, in local density (CLIP prototypes are visually dense, i.e., different classes' CLIP prototypes are close together; DINOv2 prototypes are an order of magnitude sparser), and in what kind of visual similarity they emphasize. Any fusion scheme that operates by mixing raw embeddings, rather than mixing comparable class-score outputs, risks combining signals that are not on the same scale.

Five-shot prototype noise that does not resolve with session age. Every novel-class prototype is built from only K images. On coarser benchmarks this noise correlates with how early or late a class was introduced (later sessions are, on average, slightly harder), but on fine-grained benchmarks the correlation with introduction session all but disappears, and the noise instead tracks local cluster density: a novel class that happens to sit in a visually crowded neighborhood of the embedding space is hard to represent well from five images regardless of when it was introduced. Any refinement mechanism must therefore be precision-favoring rather than session-favoring.

Label scarcity as a hard constraint on refinement. The only source of additional information beyond the K-shot support set is the unlabeled query batch, which is contaminated by base-class queries and, when the model is wrong, by mislabeled novel-class queries. Any mechanism that uses this batch to refine prototypes must be conservative enough to avoid absorbing incorrect evidence, which argues against fully transductive or hard-pseudo-label update rules that use every query indiscriminately.

Inter-class semantic confusability. On datasets that contain groups of visually and semantically similar categories (e.g., the large carnivores of CIFAR-100, the sighthound

breeds of *mini*ImageNet, or the gull species of CUB-200), the class-level semantic evidence extracted from a large language model or a simple class-name prompt can itself be insufficiently discriminative: two classes may receive semantic prototypes that are nearly identical, and any classifier built on top of them—however well the visual branch is calibrated—inherits that ambiguity whenever the visual evidence is not decisive on its own. Any correction must also avoid the opposite failure: pushing one prototype away from its closest rival must not push it into a third class.

1.4 Contributions

This thesis makes the following contributions.

1. **A tri-level reformulation of frozen FSCIL.** We present, with full derivations and worked intuition, the TRiC framework, which separates the assembly of a frozen FSCIL classifier into semantic calibration, inter-model visual calibration, and inter-modal calibration, and we discuss the empirical evidence that this separation is necessary once a second, structurally different visual encoder is introduced alongside a vision-language encoder.

2. **A label-free query-side rectification stage.** We present Query Prototype Rectification (QPR), which uses a precision-favoring, posterior-driven gate to select a high-purity subset of the unlabeled query batch and uses only that subset to refine noisy novel-class visual prototypes, with no additional annotation.

3. **A label-only semantic rerouting stage (the principal contribution of this thesis).** We design Discriminative Prompt Rerouting (DPR). DPR builds a directional pairwise discriminative prompt matrix in which every entry holds three prompts that each contrast a class with one rival through a different attribute. It then repeatedly selects the most confusable unvisited pair from the semantic similarity matrix, forms a similarity-weighted pull toward the discriminative prompt embeddings, and accepts each side's rerouted prototype only if it reduces the pair's confusion and does not raise its similarity to any other class by more than a fixed margin. We give a complete algorithmic specification and a termination and complexity argument, illustrated with real class pairs.

4. **A thorough empirical analysis of the validated backbone.** Drawing on the evaluation of TRiC and TRiC-QPR, we present session-wise accuracy trends, a Full/Base/Novel decomposition of QPR's effect, and gate-purity diagnostics, across *mini*ImageNet, CIFAR-100, and CUB-200.

5. **An empirical evaluation of DPR.** On CIFAR-100, a cumulative ablation shows that DPR raises the average session accuracy of the DINOv2/CLIP fusion backbone by 0.37 points, lowers performance degradation by 0.47 points, and improves every incremental session. A prototype-level visualization of a nine-class CIFAR-100 subset shows how the rerouted semantic prototypes shift the decision boundaries, raising novel-class accuracy and reducing the fraction of novel queries absorbed by base classes.

Chapter 2

Related Works

2.1 Few-Shot Class-Incremental Learning

FSCIL was formalized by Tao, Hong, Chang, *et al.* [23] as class-incremental learning [8], [18] in which every incremental session is also a few-shot problem [4], [10], [22], [24]. Its central difficulty is catastrophic forgetting. Early continual-learning remedies, such as distillation against a previous model and exemplar rehearsal [14], [20], were designed for the many-shot setting.

Most FSCIL methods keep a trainable backbone and control how it evolves. Continually evolved classifiers [28] adapt the classifiers of all sessions jointly with a graph model, semantic-aware distillation [6] preserves word-embedding relations among classes, forward-compatible training [32] reserves embedding space for future classes, MetaFSCIL [7] meta-learns the update rule, and class-aware bilateral distillation [30] distills from both the base-session and previous-session models. All of them still train at every incremental session, which costs compute and can reintroduce forgetting.

2.2 Frozen FSCIL and Vision-Language Adaptation

Training-free FSCIL removes incremental updates altogether. TEEN [27] shifts each novel prototype toward similar base prototypes, and FeCAM [11] replaces the Euclidean distance with a covariance-aware Mahalanobis distance. Both operate in a single visual feature space. CLIP [19] opens a second route, because its joint image-text space allows a prototype to be built from a class name alone. A frozen CLIP model can be adapted cheaply with learned prompts [12], [31], training-free adapters [29], or test-time prompt tuning [21], but none of these methods is designed for session-

structured FSCIL.

BiMC [5] brings CLIP into frozen FSCIL through bi-level modality calibration. An LLM-enriched semantic prototype is combined with a CLIP visual prototype through a soft base-prototype shift, and the result is fused with a base-covariance and description-nearest-neighbor ensemble. BiMC is the direct predecessor of this work, and its two-level structure assumes a single visual encoder, which is exactly what breaks once DINOv2 is added.

2.3 Self-Supervised Visual Features and Transduction

DINO [2] trains a Vision Transformer [9] by self-distillation without labels, and DINOv2 [17] scales this recipe into general-purpose visual features that are strong for fine-grained recognition. Like other self-supervised objectives such as contrastive instance discrimination [3], self-distillation never aligns images with text, unlike the image-text contrastive training of CLIP [19]. DINOv2's embedding geometry therefore differs from CLIP's, which motivates the inter-model calibration of Chapter 3.

A separate line of work uses unlabeled test queries as supervision. Label propagation [16] and transductive information maximization [1] classify a query set jointly, TENT [26] minimizes prediction entropy at test time, and source-free adaptation [15] trains on the model's own predictions. In frozen FSCIL the query batch also contains base-class and misclassified queries, so an unconstrained transductive update risks absorbing wrong evidence. This is why QPR (Section 3.4) admits only queries on which the calibrated posterior is already confident.

2.4 Prompt-Based Discriminative Supervision and the Motivation for DPR

Almost all of the prompting and description-based methods discussed above build each class's semantic representation independently of every other class: a class prompt or an LLM-generated description is written and embedded on its own, with no explicit mechanism to check whether the resulting semantic prototype is actually distinguishable from that of a visually or semantically similar rival class. This is a reasonable simplification when the classes involved are coarsely different (e.g., a vehicle versus an animal), but it becomes a liability when a label space contains groups of close relatives that differ in subtle visual and linguistic details, such as two gull species that dif-

fer mainly in leg color and bill markings, or two sighthound breeds that differ mainly in ear carriage and coat texture. Semantic-aware distillation [6] does exploit relations among word vectors, but it uses them to regularize a trainable model rather than to correct the text prototypes themselves. To the best of our knowledge, no method reviewed above explicitly identifies confusable class pairs after prototypes have already been computed and then uses prompts written specifically to contrast that pair to move their semantic prototypes apart; this is precisely the gap that motivates Discriminative Prompt Rerouting (DPR), presented in Section 3.5.

Chapter 3

Proposed Methodology

This chapter presents the complete methodology studied in this thesis. Section 3.1 fixes notation. Section 3.2 defines the multi-modal prototypes that every subsequent calibration step operates on. Section 3.3 presents the three levels of TRiC. Section 3.4 presents QPR. Section 3.5 presents DPR, the principal methodological contribution of this thesis, in full detail. Section 3.6 discusses how the three mechanisms compose into a single deployable pipeline.

3.1 Problem Setting

We follow the standard FSCIL protocol [23]. A stream of sessions $\{\mathcal{D}^{(0)}, \mathcal{D}^{(1)}, \dots, \mathcal{D}^{(T)}\}$ has mutually disjoint class sets $\{\mathcal{C}^{(0)}, \mathcal{C}^{(1)}, \dots, \mathcal{C}^{(T)}\}$. The base session $\mathcal{D}^{(0)}$ contains many labeled images per class, while each incremental session $\mathcal{D}^{(t)}$, $t \geq 1$, contributes N new classes with only K labeled support images per class. At session t the observed label space, the (fixed) base-class set, and the accumulated novel-class set are

$$\mathcal{C}_{\leq t} = \bigcup_{s=0}^{t} \mathcal{C}^{(s)}, \qquad \mathcal{C}_b = \mathcal{C}^{(0)}, \qquad \mathcal{C}_n = \mathcal{C}_{\leq t} \setminus \mathcal{C}_b. \tag{3.1}$$

Thus $\mathcal{C}_n$ denotes all incremental classes observed up to session t, not only the classes introduced in the most recent session. The support set of class c is $\mathcal{X}_c = \{\mathbf{x}_{c,j}\}_{j=1}^{m_c}$, where m_c is large for $c \in \mathcal{C}_b$ and $m_c = K$ for $c \in \mathcal{C}_n$. Training images from previous sessions are never revisited; after session t the model must classify an unlabeled query batch $\mathcal{U}_t$ whose true labels lie in $\mathcal{C}_{\leq t}$, with no further training and no further annotation. Table 3.1 summarizes the notation used throughout this chapter.

Table 3.1: Summary of notation used in the proposed methodology.

Symbol	Meaning
f_I, f_T, f_D	Frozen CLIP image encoder, CLIP text encoder, DINOv2 image encoder
$q_C(\mathbf{x})$, $q_D(\mathbf{x})$	Normalized CLIP-image and DINOv2 features of image $\mathbf{x}$
$v_{C,c}$, $v_{D,c}$	CLIP and DINOv2 visual prototypes of class c
w_c, $\bar{z}_c$, s_c	Prompt embedding, description mean, and Level-1 semantic prototype of class c
λ_T, ω, β, ρ	Semantic-, inter-model-, inter-modal-mixing coefficients, and softmax scale (TRiC)
τ_q, $m_{\min}$, α	Admission threshold, minimum pool size, interpolation weight (QPR)
M, k, $N_{i,j}$	Pairwise discriminative prompt matrix, prompts per pair, discriminative prototype (DPR)
S, θ, γ, ϵ	Semantic similarity matrix, pair threshold, step size, guard margin (DPR)

3.2 Multi-Modal Prototypes

Encoder notation. Let f_I and f_T be the frozen CLIP image and text encoders, and let f_D be the frozen DINOv2 image encoder. For any nonzero vector $\mathbf{u}$, $\mathrm{norm}(\mathbf{u})$ denotes ℓ_2 normalization. We use unit-normalized query and support features throughout: for any image $\mathbf{x}$,

$$\mathrm{norm}(\mathbf{u}) = \frac{\mathbf{u}}{\|\mathbf{u}\|_2}, \qquad q_C(\mathbf{x}) = \mathrm{norm}\big(f_I(\mathbf{x})\big), \qquad q_D(\mathbf{x}) = \mathrm{norm}\big(f_D(\mathbf{x})\big). \tag{3.2}$$

Semantic inputs. For each class $c \in \mathcal{C}_{\le t}$, we use a class prompt π_c (e.g., "an image of a [class name]") and n_c description sentences $\{d_{c,r}\}_{r=1}^{n_c}$, drawn from a fixed prompt template and a pool of short, factual descriptions of the class produced once, offline, by an LLM. The prompt embedding w_c, each description embedding $z_{c,r}$, and the normalized description mean $\bar{z}_c$ are

$$w_c = \mathrm{norm}\big(f_T(\pi_c)\big), \qquad z_{c,r} = \mathrm{norm}\big(f_T(d_{c,r})\big), \qquad \bar{z}_c = \mathrm{norm}\Big(\frac{1}{n_c}\sum_{r=1}^{n_c} z_{c,r}\Big). \tag{3.3}$$

Visual prototypes. For each observed class we build one CLIP visual prototype and one DINOv2 visual prototype from the same support set $\mathcal{X}_c$:

$$v_{C,c} = \mathrm{norm}\Big(\frac{1}{m_c}\sum_{j=1}^{m_c} q_C(\mathbf{x}_{c,j})\Big), \qquad v_{D,c} = \mathrm{norm}\Big(\frac{1}{m_c}\sum_{j=1}^{m_c} q_D(\mathbf{x}_{c,j})\Big), \qquad c \in \mathcal{C}_{\le t}. \tag{3.4}$$

All four quantities—w_c, $\bar{z}_c$, $v_{C,c}$, $v_{D,c}$—are computed once per class and cached; no gradient ever flows through f_I, f_T, or f_D.

3.3 Tri-Level Calibration (TRiC)

TRiC assembles the final classifier in three explicit levels, each responsible for calibrating one specific source of incompatibility. Figure 3.1 shows the three levels and how their outputs flow into the calibrated classifier.

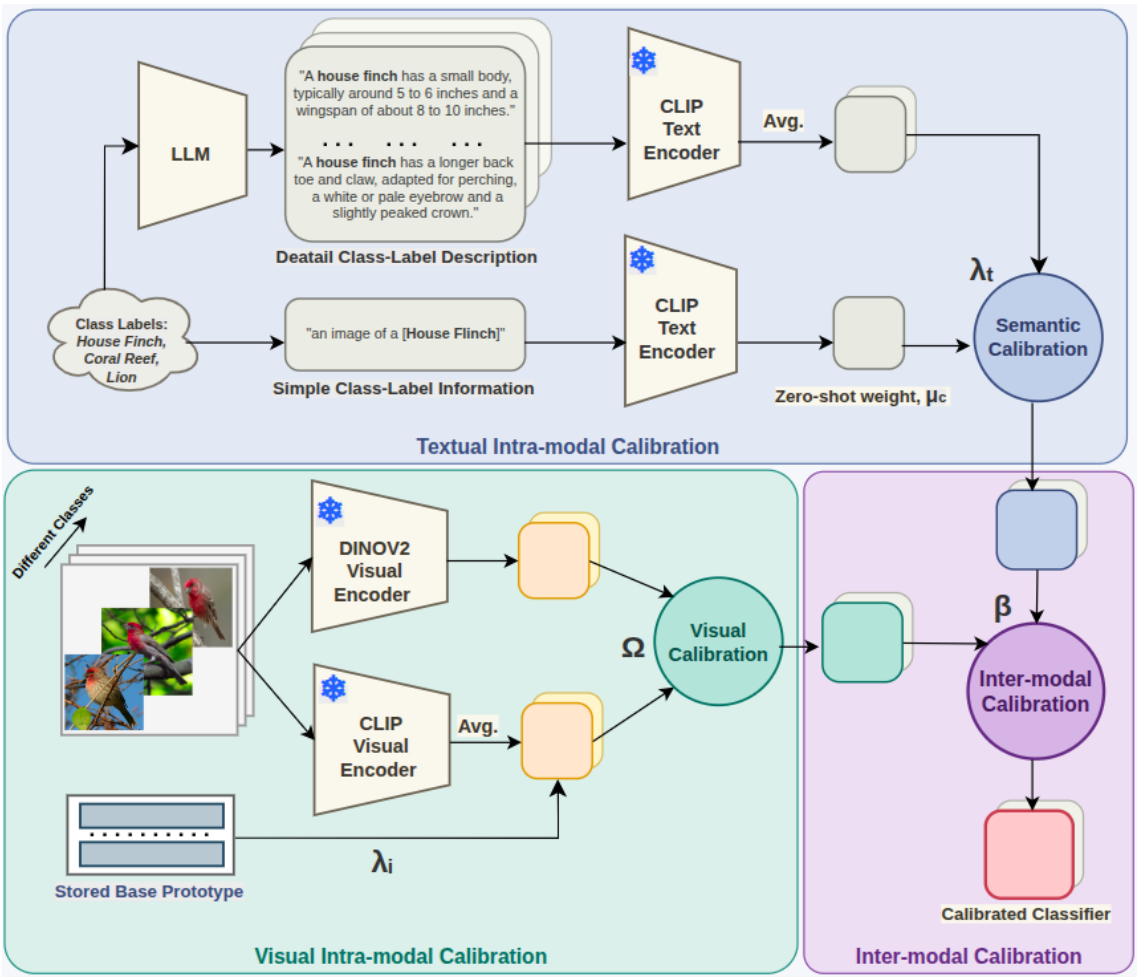


Figure 3.1: Tri-Level Calibration (TRiC). *Top:* textual intra-modal calibration (Level 1) mixes the zero-shot prompt weight μ_c (the prompt embedding w_c of Eq. 3.3) with the averaged embedding of LLM-generated class descriptions through λ_t (λ_T in Eq. 3.5). *Bottom left:* visual calibration (Level 2) combines DINOv2 and CLIP visual evidence with weight Ω (ω in Eq. 3.7); it is intra-modal because both encoders are visual, and inter-model because it fuses two encoders. The stored-base-prototype path weighted by λ_i is BiMC's soft base-prototype shift of the CLIP visual prototype, which the deployed configuration disables (Section 4.2). *Bottom right:* inter-modal calibration (Level 3) fuses the semantic and visual evidence with weight β (Eq. 3.8) into the calibrated classifier.

3.3.1 Level 1: Semantic Calibration

A class prompt embedding w_c is cheap to obtain but generic; a description mean $\bar{z}_c$ is more specific but is only as reliable as the description pool. Level 1 combines the two, before any visual evidence is used, with a single interpolation coefficient $\lambda_T \in [0, 1]$ that controls how much description information enters the final semantic prototype s_c:

$$s_c = \text{norm}\big((1 - \lambda_T)\, w_c + \lambda_T\, \bar{z}_c\big), \qquad 0 \le \lambda_T \le 1. \tag{3.5}$$

For a query image $\mathbf{x}$, the semantic score of class c is the cosine similarity between the CLIP image embedding and the semantic prototype, which is well defined precisely because CLIP's joint image-text training aligns both in one space:

$$\ell_c^S(\mathbf{x}) = q_C(\mathbf{x})^\top s_c, \qquad c \in \mathcal{C}_{\le t}. \tag{3.6}$$

Level 1 builds each s_c from class c's own prompt and descriptions only. It never compares s_c with the semantic prototype of any other class. Section 3.5 returns to this property.

3.3.2 Level 2: Inter-Model Visual Calibration

The CLIP-image and DINOv2 branches first produce separate class-score vectors—vectors of cosine similarities indexed by class—and are fused along this common axis rather than averaged in either encoder's raw embedding space. This is the key structural change relative to a bi-level (single-encoder) design: because $\ell_c^C(\mathbf{x})$ and $\ell_c^D(\mathbf{x})$ are both bounded cosine similarities in $[-1, 1]$, they are already on a comparable scale and can be fused directly, with no per-branch rescaling required beforehand:

$$\ell_c^C(\mathbf{x}) = q_C(\mathbf{x})^\top v_{C,c}, \qquad \ell_c^D(\mathbf{x}) = q_D(\mathbf{x})^\top v_{D,c}, \qquad \ell_c^V(\mathbf{x}) = \omega\, \ell_c^D(\mathbf{x}) + (1-\omega)\, \ell_c^C(\mathbf{x}), \quad 0 \le \omega \le 1, \tag{3.7}$$

where $\ell_c^V(\mathbf{x})$ is the fused visual logit and the coefficient ω weights DINOv2 evidence against CLIP visual evidence. BiMC [5] additionally applies a soft base-prototype shift to the CLIP visual prototype, drawn as the λ_i path in Figure 3.1. TRiC drops this shift in its deployed configuration because it injects base-class bias into the visual evidence that DINOv2 is meant to carry.

3.3.3 Level 3: Inter-Modal Calibration

Semantic and visual evidence are fused at the logit level, and the result is converted into a proper posterior over $\mathcal{C}_{\leq t}$ with a positive softmax scale ρ:

$$\ell_c^F(\mathbf{x}) = \beta\, \ell_c^S(\mathbf{x}) + (1-\beta)\, \ell_c^V(\mathbf{x}), \qquad p_{\text{TRiC}}(c \mid \mathbf{x}) = \frac{\exp\left(\rho\, \ell_c^F(\mathbf{x})\right)}{\sum_{k \in \mathcal{C}_{\leq t}} \exp\left(\rho\, \ell_k^F(\mathbf{x})\right)}, \qquad 0 \leq \beta \leq 1,\ \rho > 0, \tag{3.8}$$

where β weights semantic evidence against visual evidence in the final decision. The classifier used to report accuracy at every session is $\hat{y}(\mathbf{x}) = \arg\max_c p_{\text{TRiC}}(c \mid \mathbf{x})$.

Because Level 2 is entirely visual and Level 3 is entirely a function of already-computed class scores, introducing DINOv2 costs one additional interpolation coefficient (ω) rather than a full redesign of the fusion rule, and the semantic branch (Level 1) is left completely untouched by the presence of a second visual encoder. This is exactly the property that makes it possible to insert DPR (Section 3.5) between Level 1 and Level 3 without altering Level 2 or the form of Level 3 at all.

3.4 Query Prototype Rectification (QPR)

After Level 3 produces the calibrated posterior $p^*(c \mid \mathbf{x}) := p_{\text{TRiC}}(c \mid \mathbf{x})$, every novel-class DINOv2 prototype $v_{D,c}$, $c \in \mathcal{C}_n$, is still estimated from only K images. The unlabeled query batch $\mathcal{U}_t$ contains additional structure, but it also contains base-class queries and, whenever the classifier is wrong, misclassified novel-class queries. QPR therefore uses p^* only to select a high-precision subset of $\mathcal{U}_t$ and updates only novel DINOv2 prototypes, leaving semantic prototypes, CLIP visual prototypes, and base DINOv2 prototypes entirely untouched. Figure 3.2 illustrates the module on the novel class *coral reef*.

Gate. For each query $\mathbf{x} \in \mathcal{U}_t$, define its overall prediction $\hat{y}(\mathbf{x})$, its best novel-class prediction $\hat{y}_n(\mathbf{x})$, and its total novel posterior mass $m_n(\mathbf{x})$:

$$\hat{y}(\mathbf{x}) = \arg\max_{k \in \mathcal{C}_{\leq t}} p^*(k \mid \mathbf{x}), \qquad \hat{y}_n(\mathbf{x}) = \arg\max_{k \in \mathcal{C}_n} p^*(k \mid \mathbf{x}), \qquad m_n(\mathbf{x}) = \sum_{k \in \mathcal{C}_n} p^*(k \mid \mathbf{x}). \tag{3.9}$$

A query is admitted into the global gate $\mathcal{G}$ only if its top-class prediction is already novel and its total novel posterior mass exceeds an admission threshold $\tau_q \in [0, 1]$;

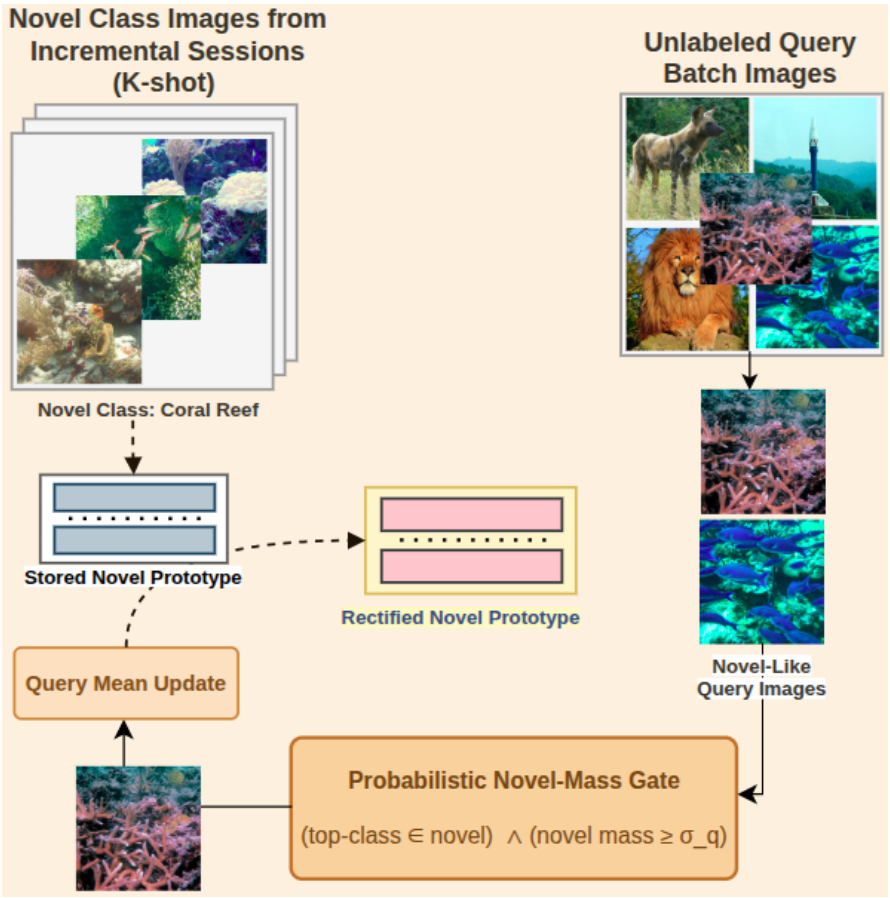


Figure 3.2: Query Prototype Rectification (QPR), illustrated with the novel class *coral reef*. Unlabeled query images pass the probabilistic novel-mass gate only if their top class is novel and their total novel posterior mass reaches the admission threshold (σ_q in the figure, τ_q in Eq. 3.10). The admitted novel-like queries form a probability-weighted query mean (Eq. 3.11), which is interpolated with the stored K-shot novel prototype to give the rectified novel prototype (Eq. 3.12).

admitted queries are then routed to a class-specific pool $\mathcal{G}_c$:

$$\mathcal{G} = \{\mathbf{x} \in \mathcal{U}_t \,:\, \hat{y}(\mathbf{x}) \in \mathcal{C}_n \text{ and } m_n(\mathbf{x}) \geq \tau_q\}, \qquad \mathcal{G}_c = \{\mathbf{x} \in \mathcal{G} \,:\, \hat{y}_n(\mathbf{x}) = c\}, \qquad c \in \mathcal{C}_n. \tag{3.10}$$

If $|\mathcal{G}_c| < m_{\min}$ for a minimum pool size $m_{\min} \in \mathbb{N}$, class c is judged not to have enough high-confidence evidence, and its prototype is left unchanged at $v_{D,c}$.

Update. For a novel class with a sufficiently large admitted pool, a probability-weighted DINOv2 query mean is computed, using each admitted query's own novel-class posterior as its weight:

$$a_c(\mathbf{x}) = \frac{p^*(c \mid \mathbf{x})}{\sum_{\mathbf{x}' \in \mathcal{G}_c} p^*(c \mid \mathbf{x}')}, \qquad \tilde{v}_{D,c} = \operatorname{norm}\Big(\sum_{\mathbf{x} \in \mathcal{G}_c} a_c(\mathbf{x})\, q_D(\mathbf{x}) \Big), \qquad \sum_{\mathbf{x} \in \mathcal{G}_c} a_c(\mathbf{x}) = 1. \tag{3.11}$$

The rectified DINOv2 prototype is a convex interpolation between the original K-shot support prototype and the query mean, controlled by $\alpha \in [0, 1]$; every other prototype is left unchanged:

$$v'_{D,c} = \begin{cases} \operatorname{norm}\big((1-\alpha)\, v_{D,c} + \alpha\, \tilde{v}_{D,c}\big), & c \in \mathcal{C}_n \text{ and } |\mathcal{G}_c| \geq m_{\min}, \\ v_{D,c}, & \text{otherwise.} \end{cases} \tag{3.12}$$

The rectified classifier is obtained by substituting $v'_{D,c}$ for $v_{D,c}$ in Equation 3.7 and recomputing Equation 3.8. Every encoder and every other prototype remains frozen; QPR only ever touches the small subset of novel DINOv2 prototypes that both attract enough high-confidence queries and are themselves the object of a confident novel-class prediction.

3.5 Discriminative Prompt Rerouting (DPR)

QPR repairs residual novel-visual-prototype noise using unlabeled query evidence. It does nothing for the semantic branch, and it does nothing at all for base classes, which never pass through its gate. This section presents Discriminative Prompt Rerouting (DPR), the methodological contribution of this thesis. DPR is a label-only refinement of the Level-1 semantic prototypes $\{s_c\}_{c \in \mathcal{C}_{\leq t}}$. It requires no images, novel or otherwise, and it reduces semantic confusion between pairs of classes whose prototypes are already highly similar before any visual evidence is considered.

3.5.1 Pairwise Discriminative Prompt Matrix

Let $T_c \equiv s_c$ denote the Level-1 semantic prototype of class c (we keep the symbol T_c in this section to make the rerouting equations self-contained). Because every T_c is unit-normalized, the pairwise semantic similarity between classes i and j is simply their cosine similarity,

$$S_{i,j} = T_i^\top T_j, \qquad i \neq j, \qquad S_{i,j} = S_{j,i}. \tag{3.13}$$

DPR draws its discriminative evidence from a pairwise discriminative prompt matrix M of shape $n \times n \times k$, where n is the number of classes it covers and k is the number of prompts per pair:

$$M_{i,j} = \left(\pi_{i,j,1}, \dots, \pi_{i,j,k}\right) \quad \text{for } i \neq j, \qquad M_{i,i} = \varnothing. \tag{3.14}$$

Entry $M_{i,j}$ holds k prompts that describe the *anchor* class i while contrasting it against the *rival* class j. We use $k = 3$ throughout. The matrix has three structural properties.

- **It is directional.** $M_{i,j} \neq M_{j,i}$: the prompts of $M_{i,j}$ describe class i and name class j only as the point of contrast, while $M_{j,i}$ does the reverse.
- **Its diagonal is empty.** A class is never contrasted with itself.
- **Each prompt of a pair targets a different attribute.** For the Saluki against the Ibizan hound, one prompt contrasts ear carriage, one contrasts coat texture, and one contrasts nose and eye color. Averaging their embeddings therefore spans several discriminative cues instead of repeating one.

Every prompt follows the template "a photo of a {class i}, {attribute of i} {contrast with class j}". The prompts are generated once, offline, by an LLM from the list of class names, exactly as the description pool $\{d_{c,r}\}$ of Equation 3.3 is generated, so building M requires no training image. Figure 3.3 shows the sub-blocks of M for two dog and two gull classes, and Appendix B lists every prompt used in this thesis.

Passing all k prompts of the pair (i, j) through the frozen CLIP text encoder and averaging gives the pairwise discriminative prototype of class i relative to class j:

$$N_{i,j} = \operatorname{norm}\Big(\frac{1}{k}\sum_{r=1}^{k} f_T(\pi_{i,j,r})\Big). \tag{3.15}$$

Each class therefore receives up to $n - 1$ discriminative prototypes, one against each rival, and the matrix yields $n(n-1)$ directed prototypes in total. Because Equation 3.15

(a) *mini*ImageNet: sub-block of M for Saluki and Ibizan hound



(b) CUB-200: sub-block of M for Ring-billed Gull and Herring Gull



Figure 3.3: Two 2×2 sub-blocks of the pairwise discriminative prompt matrix M, for (a) Saluki and Ibizan hound in *mini*ImageNet and (b) Ring-billed Gull and Herring Gull in CUB-200. Rows are anchor classes i and columns are rival classes j. Each off-diagonal cell lists the $k = 3$ prompts of $M_{i,j}$, with the attribute that each prompt contrasts shown in bold; the diagonal is empty. The matrix is directional: the Saluki row describes the Saluki against the Ibizan hound, and the Ibizan hound row describes the Ibizan hound against the Saluki. Passing the three prompts of a cell through the frozen CLIP text encoder and averaging gives the discriminative prototype $N_{i,j}$ of Equation 3.15. The full $5 \times 5 \times 3$ matrices are listed in Appendix B.

depends only on class names and prompts, all of them can be pre-computed once, offline, in parallel, with no dependency on any support or query image.

In the general design, M covers every ordered pair of observed classes. DPR, however, only ever acts on pairs whose similarity exceeds the threshold θ (Section 3.5.2), and such pairs occur inside groups of close relatives. In our implementation we therefore populate M for the most confusable group of each benchmark (Table 4.2), and the sums over rival classes in Section 3.5.2 run over the populated entries.

3.5.2 Iterative Rerouting Algorithm

DPR proceeds in rounds. At each round it identifies the single most confusable, not-yet-visited pair of classes, proposes to reroute both of their semantic prototypes away from each other using Equation 3.15, and accepts each side of the proposal independently, subject to two guards. Algorithm 1 gives the complete procedure.

Selection. Maintain a set $\mathcal{V}$ of already-visited unordered class pairs, initially empty. At each round, let

$$(a, b) = \arg \max_{\{i,j\} \notin \mathcal{V},\ i \neq j} S_{i,j}. \tag{3.16}$$

If $S_{a,b}$ does not exceed a fixed threshold θ (we use $\theta = 0.5$), or if every pair has already been visited, the algorithm terminates: no remaining pair is confusable enough to justify rerouting. Ties in Equation 3.16 are broken arbitrarily. Each unordered pair is rerouted at most once, so $\{a, b\}$ is added to $\mathcal{V}$ as soon as it is selected.

Pull vectors and candidates. DPR builds a discrimination-weighted pull vector for each side of the pair. It aggregates the discriminative prototypes of a (respectively b) relative to every rival class, weighted by how similar a (respectively b) currently is to that rival:

$$D_{a \to b} = \sum_{i \neq a} S_{a,i}\, N_{a,i}, \qquad D_{b \to a} = \sum_{i \neq b} S_{b,i}\, N_{b,i}. \tag{3.17}$$

Because $S_{a,b}$ is the largest similarity in the matrix when the pair is selected, the term $N_{a,b}$—the prompts that contrast a with b—carries the largest weight in $D_{a \to b}$. The remaining terms keep the pull informed by how class a relates to its other close relatives, not only to b. A candidate pair of rerouted prototypes is then formed with a step size $\gamma > 0$:

$$T'_a = \mathrm{norm}\left(T_a + \gamma\, D_{a \to b}\right), \qquad T'_b = \mathrm{norm}\left(T_b + \gamma\, D_{b \to a}\right). \tag{3.18}$$

Temporary similarities. The candidates are not accepted unconditionally. DPR

first computes the similarity of each candidate to the other candidate and to every remaining class:

$$\text{tmp}_{a',b'} = T_a'^{\top} T_b', \qquad \text{tmp}_{a',i} = T_a'^{\top} T_i, \qquad \text{tmp}_{b',i} = T_b'^{\top} T_i, \qquad i \notin \{a, b\}. \tag{3.19}$$

Guard 1: the pair must become less confusable. The round continues only if the rerouting actually reduced the confusion between a and b:

$$S_{a,b} > \text{tmp}_{a',b'}. \tag{3.20}$$

If Equation 3.20 fails, both prototypes are kept and the round ends.

Guard 2: no other class may become much more confusable. If Equation 3.20 holds, each side is tested independently against collateral damage to its separation from *every* other class, using a fixed margin ϵ (we use $\epsilon = 0.1$):

$$S_{a,i} - \text{tmp}_{a',i} > -\epsilon \quad \text{for all } i \notin \{a, b\} \quad \Rightarrow \quad T_a \leftarrow T_a', \tag{3.21}$$

$$S_{b,i} - \text{tmp}_{b',i} > -\epsilon \quad \text{for all } i \notin \{a, b\} \quad \Rightarrow \quad T_b \leftarrow T_b'. \tag{3.22}$$

Equation 3.21 reads: after rerouting, the similarity of class a to any single remaining class may rise by less than ϵ. The condition is a conjunction over all $n - 2$ remaining classes, so a single class whose similarity would rise by ϵ or more is enough to reject the update. Each side is accepted or rejected on its own; only T_a, only T_b, both, or neither may be updated in a given round.

Refresh. Whenever a prototype is accepted, its row and column of S are recomputed from the current prototypes before the next round begins, so that the next selection sees the rerouted geometry. The algorithm then returns to the selection step and repeats until termination.

3.5.3 Termination and Complexity

Every iteration of the loop in Algorithm 1 adds exactly one previously unvisited pair to $\mathcal{V}$, and there are only $\binom{n}{2}$ unordered pairs among n classes, so the loop terminates after at most $\binom{n}{2}$ rounds even if the threshold check never triggers first. Each round performs $O(n)$ vector operations to build $D_{a\to b}$ and $D_{b\to a}$, $O(n)$ operations to compute the temporary similarities, $O(n)$ comparisons for Guard 2, and $O(n)$ operations to refresh the affected rows of S. The total cost of Algorithm 1 is therefore $O(n^3)$ vector operations in the worst case, plus a one-time cost of $n(n-1)k$ text-encoder passes

Algorithm 1 Discriminative Prompt Rerouting (DPR)

Require: semantic prototypes $\{T_c\}_{c \in \mathcal{C}_{\le t}}$, discriminative prototypes $\{N_{i,j}\}$, threshold θ, margin ϵ, step size γ
Ensure: rerouted semantic prototypes $\{T_c\}_{c \in \mathcal{C}_{\le t}}$

$\mathcal{V} \leftarrow \varnothing$
$S_{i,j} \leftarrow T_i^\top T_j$ for all $i \neq j$
loop
  $(a, b) \leftarrow \arg\max_{\{i,j\} \notin \mathcal{V}} S_{i,j}$
  if no such pair remains **or** $S_{a,b} \le \theta$ **then**
    break
  end if
  $\mathcal{V} \leftarrow \mathcal{V} \cup \{\{a, b\}\}$
  $D_{a \to b} \leftarrow \sum_{i \neq a} S_{a,i} N_{a,i}$; $\quad D_{b \to a} \leftarrow \sum_{i \neq b} S_{b,i} N_{b,i}$
  $T'_a \leftarrow \mathrm{norm}(T_a + \gamma D_{a \to b})$; $\quad T'_b \leftarrow \mathrm{norm}(T_b + \gamma D_{b \to a})$
  compute $\mathrm{tmp}_{a',b'}$, $\mathrm{tmp}_{a',i}$, $\mathrm{tmp}_{b',i}$ for all $i \notin \{a, b\}$ ▷ Eq. 3.19
  if $S_{a,b} > \mathrm{tmp}_{a',b'}$ **then** ▷ Guard 1
    if $S_{a,i} - \mathrm{tmp}_{a',i} > -\epsilon$ for every $i \notin \{a, b\}$ **then** ▷ Guard 2, side a
      $T_a \leftarrow T'_a$
    end if
    if $S_{b,i} - \mathrm{tmp}_{b',i} > -\epsilon$ for every $i \notin \{a, b\}$ **then** ▷ Guard 2, side b
      $T_b \leftarrow T'_b$
    end if
    recompute the rows and columns of S for every accepted side
  end if
end loop
return $\{T_c\}_{c \in \mathcal{C}_{\le t}}$

to compute every discriminative prototype $N_{i,j}$. That one-time cost is embarrassingly parallel and can be cached offline, since it does not depend on any support or query image. In our implementation each prompt matrix covers $n = 5$ classes, so DPR visits at most ten pairs per benchmark and encodes sixty prompts. Even in the general design, with $n \leq 200$ classes (Chapter 4), the cost is negligible next to a single forward pass of the CLIP or DINOv2 visual encoders over a session's images.

3.6 Integration and Overall Pipeline

The three mechanisms presented in this chapter operate on three disjoint sources of evidence, in increasing order of what they require. DPR (Section 3.5) requires only class labels and prompts; its prompt matrix is built offline, and the rerouting itself can run before a single support or query image of a session has been observed. The three levels of TRiC (Section 3.3) require the K-shot labeled support set of each new class. QPR (Section 3.4) requires the unlabeled query batch that only becomes available once a session is underway. This ordering is also a supervision-strength ordering, from least to most session-specific evidence, and the pipeline is applied in exactly this order within each session. TRiC Level 1 builds s_c for the accumulated class set $\mathcal{C}_{\leq t}$; DPR reroutes these prototypes; Levels 2 and 3 build and fuse visual and semantic evidence (Equation 3.7–Equation 3.8) with the rerouted prototypes in place of the raw Level-1 output; and QPR (Equation 3.9–Equation 3.12) then consumes the resulting calibrated posterior $p^*(c \mid \mathbf{x})$ as its gate signal, exactly as it would without DPR. Because DPR only ever replaces the input to Equation 3.6 and touches no other equation, and because QPR only ever reads $p^*(c \mid \mathbf{x})$ and writes to $v_{D,c}$, all three mechanisms compose without any structural conflict. Every encoder, and every prototype not explicitly targeted by DPR or QPR, remains exactly as frozen as in the original two-module design of Figure 3.1 and Figure 3.2.

Chapter 4

Experimental Setup

This chapter describes the datasets, protocol, implementation details, and evaluation metrics used to evaluate TRiC and TRiC-QPR, followed by the experimental setup used to evaluate DPR. The results of all experiments are reported in Chapter 5.

4.1 Datasets and Protocol

We use the three standard FSCIL benchmarks and their standard protocols [5], [23], summarized in Table 4.1.

Table 4.1: FSCIL benchmarks used in this thesis. "Way" is the number of new classes introduced per incremental session and "shot" is the number of labeled images per new class.

Dataset	Base classes	Incremental sessions	Way	Shot	Total classes
CIFAR-100 [13]	60	8	5	5	100
*mini*ImageNet [24]	60	8	5	5	100
CUB-200 [25]	100	10	10	5	200

CIFAR-100 and *mini*ImageNet use 60 base classes and eight 5-way 5-shot incremental sessions (sessions A_0 through A_8); CUB-200-2011, a fine-grained bird recognition dataset, uses 100 base classes and ten 10-way 5-shot incremental sessions (sessions A_0 through A_{10}), and is the benchmark on which fine-grained inter-class confusability is most pronounced. Following standard practice, we report accuracy at every session over all classes observed so far, averaged over 20 random seeds per configuration for the TRiC and TRiC-QPR experiments, with a fixed, non-overlapping evaluation query set drawn from every class observed so far.

4.2 Implementation Details

All variants use a frozen CLIP ViT-B/16 image and text encoder [19]; TRiC additionally uses a frozen DINOv2 ViT-B/14 image encoder [17], which is comparable in computational cost (FLOPs) to the CLIP image encoder, so the dominant inference cost of TRiC-QPR is approximately twice that of a CLIP-only frozen FSCIL method. Class descriptions and prompt templates follow BiMC [5]. For TRiC we fix $\lambda_T = 0.5$ and disable BiMC's soft visual calibration (the λ_i path of Figure 3.1), its base-covariance branch, its description-nearest-neighbor branch, and its masked auxiliary ensemble, so that a single calibrated posterior remains for QPR to consume. Unless otherwise noted, the remaining deployed defaults are: inter-model weight $\omega = 0.7$; softmax scale $\rho = 100$; inter-modal weight $\beta = 0.65$ on CIFAR-100 and *mini*ImageNet and $\beta = 0.30$ on CUB-200 (the only hyperparameter allowed to vary by dataset, following the convention of the predecessor framework [5]); and QPR defaults $\tau_q = 0.3$, $m_{\min} = 5$, $\alpha = 0.75$, fixed across all three datasets and all seeds, i.e., with no per-seed tuning. The DPR settings are given in Section 4.4.

4.3 Evaluation Metrics

We report four complementary metrics throughout Chapter 5:

1. **Session-wise accuracy** A_t, the fraction of the cumulative test set (every class observed up to and including session t) correctly classified at session t.
2. **Average accuracy**, the mean of $A_0, \dots, A_T$ across all sessions.
3. **Performance degradation**, $\mathrm{PD} = A_0 - A_T$, the drop from base-session to final-session accuracy, which isolates how much of a method's base-session advantage survives the incremental stream.
4. **Full/Base/Novel decomposition**, which separately reports final-session accuracy on the base classes $\mathcal{C}_b$, the novel classes $\mathcal{C}_n$, and their union.

For QPR specifically, we additionally report gate purity (the fraction of gated queries whose true class is indeed novel), base leakage (its complement), the number of gated queries, and the number of novel classes whose prototype was actually updated (Section 5.2). For the prototype-level analysis of DPR, we also report novel-class accuracy and the *novel→base rate*, the fraction of novel-class queries that the classifier assigns to a base class (Section 4.4.2).

4.4 Experimental Setup for DPR

DPR is evaluated in two complementary ways: a cumulative, session-wise ablation on CIFAR-100 that measures its effect on accuracy and forgetting, and a prototype-level visualization that shows how the rerouted prototypes move the decision boundaries. Both use the prompt matrices described below, with $k = 3$, $\theta = 0.5$, and $\epsilon = 0.1$.

4.4.1 Discriminative Prompt Matrix Construction

For each benchmark, the prompt matrix M of Section 3.5.1 is instantiated over the hardest confusable group of that benchmark, because DPR only activates on pairs whose semantic similarity exceeds θ. Table 4.2 lists the three groups. For every ordered pair (i, j) of distinct classes in a group, an LLM wrote $k = 3$ prompts that describe class i against class j, each targeting a different distinguishing attribute, such as coat pattern, body build, and tail shape for the large carnivores, ear carriage, coat texture, and nose color for the hounds, or leg color, bill markings, and size for the gulls. Each matrix therefore has shape $5 \times 5 \times 3$, with 20 directed pairs and 60 prompts. The prompts are generated once, offline, from the class names alone, and every one of them is listed verbatim in Appendix B.

Table 4.2: Confusable class groups covered by the discriminative prompt matrix of each benchmark. Every group has $n = 5$ classes, 20 directed pairs, and $20 \times 3 = 60$ prompts.

Dataset	Confusable group	Classes in the prompt matrix
CIFAR-100	"large carnivores" superclass	bear, leopard, lion, tiger, wolf
*mini*ImageNet	hound and gundog breeds (training split)	Walker hound, Ibizan hound, Saluki, Gordon setter, golden retriever
CUB-200	gull species (classes 059–064)	California Gull, Glaucous-winged Gull, Heermann's Gull, Herring Gull, Ring-billed Gull

4.4.2 Prototype-Level Rerouting Visualization

To show how DPR changes the geometry of the classifier, Figure 4.1 projects a nine-class CIFAR-100 subset with t-SNE. The subset has five base classes (otter, leopard, lion, bear, and elephant) and four novel classes (porcupine, raccoon, tiger, and wolf), so it contains the whole CIFAR-100 prompt group of Table 4.2. Panels (b) and (c) share the same queries, visual prototypes, and QPR updates, so their difference isolates DPR. Rerouting raises the accuracy on the subset from 88.8% to 89.6% and the

novel-class accuracy from 87.5% to 88.8%, and it lowers the novel→base rate from 10.5% to 9.5%.

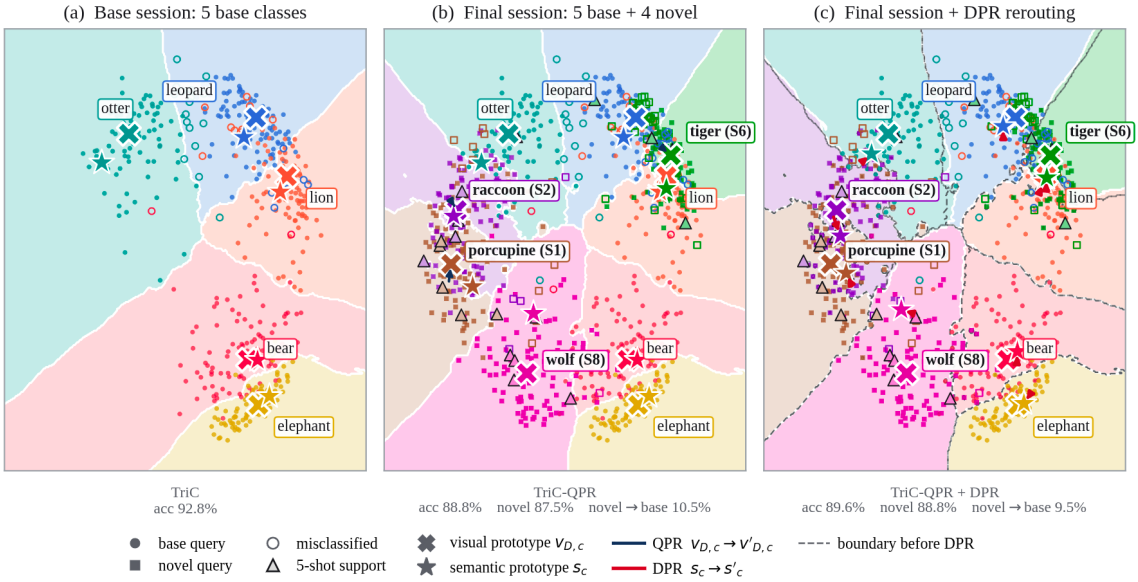


Figure 4.1: Prototype-level visualization of DPR on a nine-class CIFAR-100 subset (t-SNE). (a) Base session under TRiC. (b) Final session under TRiC-QPR, with novel classes labeled by the session that introduced them. (c) The same final session after DPR. Crosses are visual prototypes and stars are semantic prototypes; navy arrows mark QPR updates, red arrows mark DPR reroutes, and dashed lines in panel (c) are the boundaries before DPR. “acc” is subset accuracy, “novel” is novel-class accuracy, and “novel→base” is the fraction of novel queries assigned to a base class.

Chapter 5

Results and Discussion

This chapter presents the empirical results of TRiC and TRiC-QPR under the evaluation protocol of Chapter 4, followed by the measured session-wise impact of Discriminative Prompt Rerouting on CIFAR-100.

5.1 Main Results

Figure 5.1 plots session-wise accuracy for every method on all three datasets. Across all three benchmarks, TRiC and TRiC-QPR dominate every baseline throughout the incremental schedule, not only at the final session, and the gap between them and the strongest prior training-free baseline (BiMC with its full auxiliary ensemble) widens rather than narrows as more novel classes accumulate. Without any incremental training, the tri-level backbone improves on the BiMC ensemble baseline's final-session accuracy by +9.83, +1.32, and +12.53 points on CIFAR-100, *mini*ImageNet, and CUB-200 respectively (73.24 → 83.07, 92.41 → 93.73, 72.76 → 85.29), and the label-free QPR stage adds a further +2.43, +1.01, and +0.88 points on top of the same backbone (83.07 → 85.50, 93.73 → 94.74, 85.29 → 86.17). These final-session numbers come from the 20-seed Full/Base/Novel study of Section 5.2. TRiC-QPR beats BiMC in all 20 evaluated seeds on every benchmark, and, as Section 5.2 shows, these gains do not come at the price of base-class confusion.

Table 5.1, Table 5.2, and Table 5.3 give the full session-by-session comparisons on *mini*ImageNet, CIFAR-100, and CUB-200. They come from the session-wise evaluation runs plotted in Figure 5.1, so their final-session columns differ slightly from the final-session Full accuracies of the 20-seed study quoted above.

Two patterns are visible across Table 5.1–Table 5.3. First, TRiC's advantage over ev-

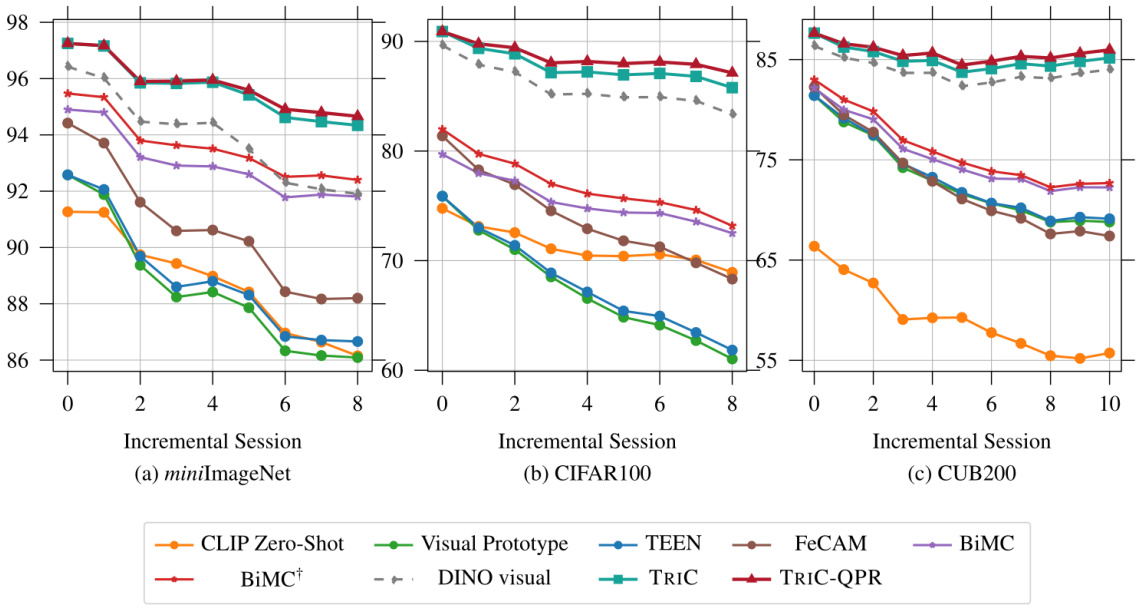


Figure 5.1: Session-wise accuracy on (a) *mini*ImageNet, (b) CIFAR-100, and (c) CUB-200. TRiC and TRiC-QPR dominate every training-free baseline throughout the incremental schedule, not only at the final session. BiMC† denotes BiMC with its auxiliary ensemble.

Table 5.1: Full session-wise comparison on *mini*ImageNet (nine sessions, A_0–A_8). V, L, and V-L denote visual, language, and vision-language modalities.

Method	Mod.	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	Avg. ↑	PD ↓
CLIP zero-shot	L	91.27	91.25	89.74	89.43	88.98	88.42	86.96	86.64	86.15	88.76	5.12
Visual Prototype (CLIP)	V	92.58	91.88	89.37	88.24	88.42	87.86	86.33	86.16	86.09	88.55	6.49
TEEN	V	92.58	92.06	89.69	88.60	88.80	88.31	86.84	86.71	86.66	88.92	5.92
FeCAM	V	94.42	93.71	91.61	90.59	90.62	90.22	88.43	88.17	88.20	90.66	6.22
BiMC	V-L	94.90	94.80	93.21	92.91	92.88	92.60	91.78	91.88	91.81	92.97	3.09
BiMC (ens.)	V-L	95.47	95.34	93.80	93.63	93.51	93.18	92.51	92.56	92.40	93.60	3.07
CLIP-visual-only	V	92.58	91.86	89.61	88.82	88.75	88.16	86.93	86.82	86.79	88.92	5.79
DINO-visual-only	V	96.42	96.02	94.47	94.38	94.43	93.49	92.30	92.07	91.90	93.94	4.52
TRiC	V-L	97.25	97.16	95.85	95.83	95.87	95.42	94.62	94.47	94.34	95.65	2.91
TRiC-QPR	V-L	**97.25**	**97.17**	**95.90**	**95.91**	**95.95**	**95.59**	**94.91**	**94.79**	**94.66**	**95.79**	**2.60**

Table 5.2: Full session-wise comparison on CIFAR-100 (nine sessions, A_0–A_8).

Method	Mod.	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	Avg. ↑	PD ↓
CLIP zero-shot	L	74.77	73.12	72.56	71.08	70.46	70.41	70.58	70.06	68.93	71.33	5.84
Visual Prototype (CLIP)	V	75.88	72.78	71.01	68.52	66.55	64.84	64.12	62.71	61.03	67.49	14.85
TEEN	V	75.88	73.00	71.39	68.88	67.14	65.41	64.94	63.44	61.84	67.99	14.04
FeCAM	V	81.38	78.28	76.93	74.55	72.91	71.81	71.26	69.80	68.32	73.92	13.06
BiMC	V-L	79.70	77.97	77.29	75.35	74.76	74.39	74.34	73.55	72.50	75.54	7.20
BiMC (ens.)	V-L	81.98	79.74	78.84	77.00	76.11	75.68	75.33	74.61	73.18	76.94	8.80
CLIP-visual-only	V	75.90	72.74	70.97	68.31	66.54	65.49	64.94	63.73	62.20	67.87	13.70
DINO-visual-only	V	89.63	87.87	87.21	85.16	85.22	84.90	84.92	84.57	83.34	85.87	6.29
TRiC	V-L	90.90	89.37	88.87	87.15	87.22	86.95	87.08	86.81	85.78	87.79	5.12
TRiC-QPR	V-L	**90.90**	**89.78**	**89.42**	**88.04**	**88.17**	**87.98**	**88.12**	**87.91**	**87.13**	**88.61**	**3.77**

Table 5.3: Full session-wise comparison on CUB-200-2011 (eleven sessions, A_0–A_{10}).

Method	Mod.	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9	A_{10}	Avg. ↑	PD ↓
CLIP zero-shot	L	66.38	64.05	62.71	59.07	59.24	59.27	57.76	56.69	55.46	55.19	55.73	59.23	10.65
Visual Prototype (CLIP)	V	81.42	78.78	77.43	74.22	72.91	71.59	70.65	69.97	68.79	68.93	68.80	73.04	12.62
TEEN	V	81.42	79.16	77.49	74.60	73.26	71.75	70.68	70.21	68.90	69.28	69.12	73.26	12.30
FeCAM	V	82.26	79.48	77.76	74.68	72.86	71.10	69.92	69.17	67.61	67.89	67.40	72.74	14.86
BiMC	V-L	82.16	79.99	79.04	76.10	75.07	74.04	73.13	73.08	71.88	72.26	72.25	75.36	9.91
BiMC (ens.)	V-L	83.00	81.01	79.83	76.96	75.84	74.73	73.86	73.47	72.26	72.61	72.68	76.02	10.32
CLIP-visual-only	V	81.63	79.14	77.49	74.24	73.26	71.68	70.81	70.03	68.53	68.72	68.47	73.09	13.16
DINO-visual-only	V	86.35	85.19	84.66	83.67	83.68	82.34	82.72	83.28	83.14	83.65	84.01	83.88	2.34
TRiC	V-L	87.64	86.25	85.80	84.83	84.91	83.74	84.10	84.59	84.35	84.82	85.17	85.12	2.47
TRiC-QPR	V-L	**87.64**	**86.58**	**86.23**	**85.40**	**85.66**	**84.46**	**84.84**	**85.32**	**85.16**	**85.62**	**85.97**	**85.72**	**1.67**

ery training-free baseline is present from the base session onward and persists through every incremental session, confirming that the gain is structural (better calibration) rather than an artifact of a particular session count. Second, QPR's contribution is concentrated in performance degradation rather than in base-session accuracy: on every dataset, A_0 is identical for TRiC and TRiC-QPR (QPR has no novel classes to rectify before any incremental session has occurred), and the entire QPR gain accumulates from A_1 onward as novel classes are introduced. On CIFAR-100, for example, QPR lowers PD from 5.12 to 3.77 points, and on CUB-200 from 2.47 to 1.67 points.

5.2 Effect of Query Prototype Rectification

Table 5.4 compares three calibrated frozen-FSCIL backbones with and without QPR, decomposed into final-session Full, Base, and Novel accuracy, at the deployed default $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$.

Table 5.4: Effect of adding QPR on top of three calibrated frozen-FSCIL backbones, at the deployed default $(\alpha, \tau_q, m_{\min}) = (0.75, 0.3, 5)$, decomposed into final-session Full/Base/Novel accuracy. The Δ rows report the per-column gain over the corresponding pre-QPR row.

		CIFAR-100			*mini*ImageNet			CUB-200		
Backbone	QPR	Full	Base	Novel	Full	Base	Novel	Full	Base	Novel
BiMC	off	72.61	75.21	68.70	91.88	93.15	89.98	72.27	80.08	64.63
BiMC	on	73.50	74.65	71.78	92.31	92.92	91.39	73.50	79.55	67.59
Δ		+0.89	−0.56	+3.08	+0.43	−0.23	+1.41	+1.23	−0.53	+2.96
BiMC + Ens.	off	73.24	77.79	66.41	92.41	92.97	91.57	72.76	78.89	66.77
BiMC + Ens.	on	73.28	76.73	68.10	92.55	92.68	92.37	73.01	77.70	68.42
Δ		+0.04	−1.06	+1.69	+0.14	−0.29	+0.80	+0.25	−1.19	+1.65
TRiC	off	83.07	88.48	74.96	93.73	96.13	90.12	85.29	87.84	82.80
TRiC	on	**85.50**	88.01	**81.75**	**94.74**	95.80	**93.15**	**86.17**	87.72	**84.66**
Δ		+2.43	−0.47	+6.79	+1.01	−0.33	+3.03	+0.88	−0.12	+1.86

QPR is a general label-free rectification stage, but its gain is set by how clean the

calibrated posterior it consumes already is. On plain BiMC, QPR adds +0.43 Full accuracy on *mini*ImageNet; on BiMC's full ensemble configuration the gain shrinks to +0.14, because BiMC's own description-nearest-neighbor branch already extracts some query-side novel evidence, leaving less for QPR to add. On TRiC, the same rectification stage delivers +1.01 Full and a sharply concentrated +3.03 Novel gain on *mini*ImageNet (and +6.79 and +1.86 Novel accuracy on CIFAR-100 and CUB-200 respectively), the largest gain of the three backbones on every dataset. This ordering supports the structural claim of Chapter 3: tri-level calibration produces the cleanest posterior for QPR's gate to act on, and the gate then targets the noisy five-shot novel prototypes precisely, without leaking that rectification into base-class accuracy, which moves by at most 0.47 points on the TRiC backbone on every dataset.

Table 5.5 reports the gate diagnostics underlying TRiC-QPR's final-session results.

Table 5.5: TRiC-QPR final-session gate diagnostics at the deployed default configuration. Gate purity is the fraction of gated queries whose true class is indeed novel; base leakage is its complement.

Dataset	Gate purity	Base leakage	Gated queries	Updated novel classes
CIFAR-100	97.79%	2.21%	3191.7 ± 27.8	40/40
*mini*ImageNet	98.57%	1.43%	3743.1 ± 25.8	40/40
CUB-200	99.38%	0.62%	2861.2 ± 8.0	99/100

Gate purity is at least 97.79% on every benchmark, base leakage never exceeds 2.21%, and the admission rule successfully accumulates enough evidence to update 40/40, 40/40, and 99/100 novel classes' prototypes on CIFAR-100, *mini*ImageNet, and CUB-200 respectively. The one CUB-200 class whose prototype pool never reaches the minimum size $m_{\min} = 5$ is a direct, quantitative illustration of the precision-over-coverage trade-off built into Equation 3.10: a conservative gate necessarily leaves some novel classes with too little admitted evidence to update safely. Appendix A shows how QPR's gains are distributed across classes.

5.3 Effect of Discriminative Prompt Rerouting

Table 5.6 reports a cumulative ablation on CIFAR-100 in which each row adds one component to the row above: BiMC's auxiliary ensemble, DINOv2 visual evidence, DINOv2/CLIP fusion, and finally DPR. The last row, Δ from DPR, is the difference between the + DPR row and the + DINOv2/CLIP fusion row, which share every other component. Because the chain keeps BiMC's ensemble, its absolute numbers are compared within the chain rather than with the TRiC rows of Table 5.2.

Table 5.6: Cumulative ablation on CIFAR-100: session-wise accuracy, average accuracy, and performance degradation (PD = $A_0 - A_8$) as BiMC's ensemble, DINOv2 evidence, DINOv2/CLIP fusion, and DPR are added in turn. The Δ row is the gain of the + DPR row over the + DINOv2/CLIP fusion row.

Variant	Accuracy in each session (%) ↑									Avg. ↑	PD ↓
	A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8		
BiMC	79.62	77.96	77.15	75.21	74.59	74.17	74.18	73.48	72.54	75.43	7.08
+ Ensemble	82.05	79.73	78.75	76.83	76.02	75.60	75.43	74.70	73.26	76.93	8.79
+ DINOv2 only	90.22	88.23	87.57	85.40	85.36	85.04	85.15	84.85	83.58	86.16	6.64
+ DINOv2/CLIP fusion	89.77	87.76	87.15	85.01	84.96	84.66	84.73	84.45	83.09	85.73	6.68
+ DPR	89.77	88.00	87.43	85.52	85.50	85.12	85.15	84.83	83.56	**86.10**	**6.21**
Δ from DPR	+0.00	+0.24	+0.28	+0.51	+0.54	+0.46	+0.42	+0.38	+0.47	+0.37	−0.47

DPR improves the fused backbone in every incremental session and leaves the base session untouched. The base-session accuracy is identical with and without DPR (89.77%, $\Delta = +0.00$), so the rerouting does not disturb the base-session classifier, and all of its gain accrues in the incremental sessions. The gain is +0.24 and +0.28 points in the first two incremental sessions, rises to between +0.46 and +0.54 points in sessions 3 to 5, and stays between +0.38 and +0.47 points through the final session. Averaged over all nine sessions, DPR raises accuracy from 85.73% to 86.10% (+0.37 points). Because A_0 is unchanged while A_8 rises from 83.09% to 83.56%, performance degradation falls from 6.68 to 6.21 points (−0.47), the lowest PD of any row in the chain. Figure 5.2(a) plots the per-session gain. Figure 5.2(b) places the same two rows against the DINOv2-only row of the chain: the fusion row trails DINOv2 alone in every session, while the DPR row closes most of that gap and rises above it in sessions 3 to 5.

The gain appears in the first incremental session, grows through the middle of the schedule, and is sustained until the end. It does not fade as more novel classes accumulate, and because the base session is unchanged, this sustained gain translates directly into less forgetting as measured by PD.

5.4 Discussion and Limitations

Several limitations of the studied system are worth stating explicitly.

- **The QPR gate is conservative but not perfect.** The criterion of Equation 3.10 checks only whether a query is novel, not which novel class it belongs to, so gate purity cannot reveal an admitted query that is routed to the wrong novel class. A gate that additionally conditions on the gap between the top two novel posteriors is a natural extension that we did not implement here.

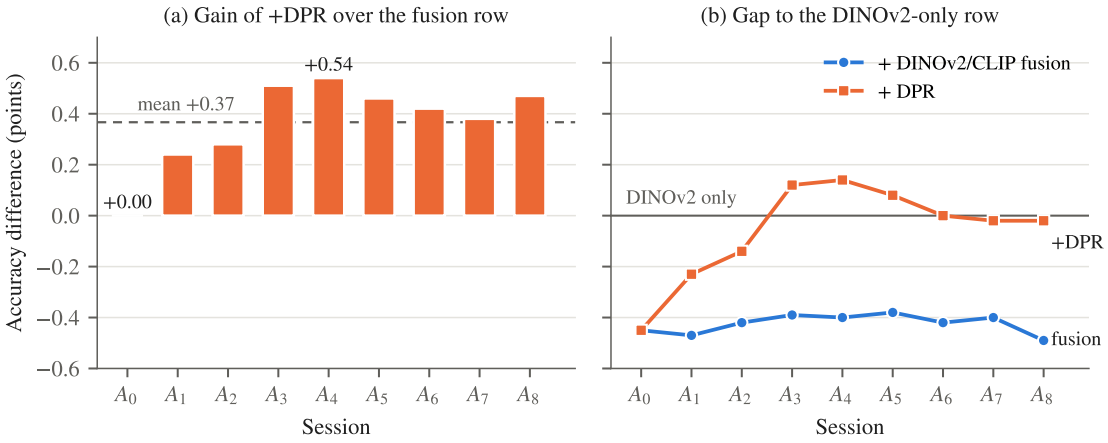


Figure 5.2: Session-wise effect of DPR on CIFAR-100, computed from Table 5.6. (a) Gain of the + DPR row over the + DINOv2/CLIP fusion row in each session; the dashed line is the mean gain over all nine sessions. (b) Gap of the fusion row and the + DPR row to the + DINOv2 only row; values above zero mean the row is more accurate than DINOv2 alone.

- **QPR is transductive.** Its update depends on the composition of the unlabeled query batch, so a session whose query batch contains very few images of a novel class leaves that class unrectified, as the one CUB-200 class below $m_{\min}$ in Table 5.5 shows.
- **Several mixing weights are fixed by hand.** Both ω (Level 2) and β (Level 3) are either fixed across all three datasets or allowed only a single, manually chosen alternate value. QPR's interpolation weight α is uniform across every novel class, so it treats a class with a large, high-purity admitted pool identically to a class that barely cleared $m_{\min}$.
- **DPR depends on the quality of its prompts.** The discriminative prompts are generated by an LLM from class names alone. A prompt that names an attribute the CLIP text encoder does not represent well, or an attribute that is wrong for the class, weakens the discriminative prototype $N_{i,j}$. The two guards prevent such a prompt from making the prototypes more confusable, but they cannot make it helpful.
- **DPR's quantitative evaluation is limited to CIFAR-100.** The session-wise ablation and the visualization both use CIFAR-100. Prompt matrices for the *mini*ImageNet hounds and the CUB-200 gulls are already built (Appendix B), but their session-wise effect is not reported here. The prompt matrix of each benchmark also covers only its single most confusable group, and the sensitivity of DPR to θ, ϵ, and γ has not been swept.

Chapter 6

Conclusion

6.1 Summary of Contributions

This thesis treated frozen, training-free FSCIL as a calibration problem with several separable levels. Tri-Level Calibration (TRiC) fuses CLIP and DINOv2 evidence on a common class-score axis, and Query Prototype Rectification (QPR) refines noisy novel DINOv2 prototypes with confidently predicted unlabeled queries. Across *mini*ImageNet, CIFAR-100, and CUB-200, TRiC improves final-session accuracy over the strongest training-free baseline by 1.32, 9.83, and 12.53 points, and QPR adds 1.01, 2.43, and 0.88 points.

Our principal contribution, Discriminative Prompt Rerouting (DPR), reroutes the semantic prototypes of the most confusable class pairs using a directional matrix of contrastive prompts, with guards that stop any other class from becoming more confusable. It uses no images and no training. On CIFAR-100, DPR improves every incremental session, raises average accuracy from 85.73% to 86.10%, and lowers performance degradation from 6.68 to 6.21 points.

6.2 Limitations

The TRiC and TRiC-QPR results in Chapter 5 are limited to three benchmarks, all of which pair a frozen CLIP ViT-B/16 with a frozen DINOv2 ViT-B/14; we do not know how the tri-level view generalizes to other encoder pairings, other backbone scales, or benchmarks outside natural-image recognition. The quantitative evaluation of DPR is limited to CIFAR-100, and its gain, while consistent across sessions, is modest in absolute size. Its prompt matrices cover only the single most confusable group of each

benchmark, and its prompts come from an LLM, so their quality bounds what DPR can achieve. QPR and DPR also depend on hyperparameters (α, τ_q, $m_{\min}$ for QPR; θ, γ, ϵ for DPR) that are either validated only in aggregate across datasets (QPR) or fixed without a sensitivity sweep (DPR).

6.3 Future Work

We identify five concrete directions for future work, in the order we believe they should be pursued.

1. **Extend the DPR evaluation.** The prompt matrices for the *mini*ImageNet hounds and the CUB-200 gulls are already built (Appendix B). Running the cumulative ablation on both benchmarks, extending each matrix beyond a single confusable group, and sweeping θ, γ, and ϵ would show how DPR scales with granularity. On CUB-200, a per-class diagnostic would measure whether DPR also reduces the number of queries routed to the wrong novel class.

2. **Make QPR's interpolation class-specific.** A per-class α for QPR, informed by admitted-pool size or posterior confidence, should be compared against the current uniform default.

3. **Select the inter-model weight per dataset.** A per-dataset, validation-selected inter-model weight ω should be compared against the current fixed default, since the benefit of DINOv2 over CLIP visual evidence need not be the same on every dataset.

4. **Add an intra-novel margin to the QPR gate.** A gate that additionally conditions on the margin between the top two novel posteriors should be designed and compared against the current gate, to catch queries that are correctly identified as novel but routed to the wrong novel class, without discarding the precision benefits already demonstrated for the existing gate.

5. **Refine novel visual prototypes with labeled evidence.** A labeled-only, intra-novel refinement of the DINOv2 novel prototype itself—for instance, a leave-one-out refinement using the remaining shots of the same class, or a margin-aware contrastive adjustment against a class's dominant DINOv2 neighbors—would fill the one part of the evidence space (labeled, novel-only) that neither TRiC, QPR, nor DPR currently addresses.

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Appendices

Appendix A

Additional Diagnostic Results

Table A.1 reports the final-session, fixed-seed distribution of per-class accuracy change after QPR is applied on top of TRiC, referenced in Section 5.2.

Table A.1: Fixed-seed, per-class final-session accuracy change after QPR, decomposed into improved, worsened, and unchanged classes.

Dataset	Classes improved	Classes worse	Classes unchanged	Total classes
CIFAR-100	39	20	41	100
*mini*ImageNet	19	11	70	100
CUB-200	30	13	157	200

The large "unchanged" fraction on every dataset is consistent with QPR's conservative gate: a class is only ever updated if its admitted pool clears $m_{\min}$, so the majority of classes—whose five-shot prototype was already reasonably accurate—are correctly left untouched, and the aggregate gains reported in Table 5.4 are carried by a minority of classes whose prototypes were initially far from their true cluster centroid.

Appendix B

Discriminative Prompt Matrices

This appendix lists every discriminative prompt used by DPR. The prompts are reproduced verbatim from the prompt matrix M described in Section 3.5.1. Entry $M_{i,j}$ holds $k = 3$ prompts that describe the anchor class i while contrasting it against the rival class j. The matrix is directional, so $M_{i,j} \neq M_{j,i}$, and diagonal entries are empty because a class is never contrasted with itself. Each of the three prompts of a pair targets a different attribute, so that their averaged text embedding $N_{i,j}$ spans several discriminative cues instead of repeating one.

B.1 CIFAR-100

The CIFAR-100 matrix covers the complete "large carnivores" superclass: bear, leopard, lion, tiger, and wolf. It has shape ($5\times5\times3$), with 20 directed pairs and 60 prompts in total (Table B.1).

Table B.1: Discriminative prompt matrix M for CIFAR-100. Each row lists the $k = 3$ prompts of $M_{i,j}$.

Anchor class i	Rival class j	Prompts in $M_{i,j}$
bear	leopard	1. "a photo of a bear, whose thick shaggy fur is a single plain color with none of the rosette spots of a leopard" 2. "a photo of a bear, with a heavy barrel shaped body far bulkier than the sleek frame of a leopard" 3. "a photo of a bear, showing a short stubby tail instead of the long trailing tail of a leopard"

Table B.1 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	lion	1. “a photo of a bear, with a broad bare faced head that never carries the thick mane of a lion” 2. “a photo of a bear, whose small rounded ears sit high on the skull unlike the flat ears of a lion” 3. “a photo of a bear, standing flat on its whole foot rather than up on its toes like a lion”
	tiger	1. “a photo of a bear, whose uniform dark coat carries none of the bold black stripes of a tiger” 2. “a photo of a bear, with a long straight snout instead of the short broad muzzle of a tiger” 3. “a photo of a bear, whose shaggy coarse fur looks nothing like the sleek orange pelt of a tiger”
	wolf	1. “a photo of a bear, with a massive heavy body many times the bulk of a slender wolf” 2. “a photo of a bear, showing small rounded ears rather than the sharp upright ears of a wolf” 3. “a photo of a bear, whose stubby tail is nothing like the long bushy tail of a wolf”
leopard	bear	1. “a photo of a leopard, its golden coat covered in dark rosette spots that are absent on a bear” 2. “a photo of a leopard, with a lithe muscular body much slimmer than a heavy bear” 3. “a photo of a leopard, whose long tail trails far behind it unlike the stub of a bear”
	lion	1. “a photo of a leopard, patterned with dark rosettes across a coat that a lion keeps plain” 2. “a photo of a leopard, with a clean maneless head unlike the shaggy ruff of a lion” 3. “a photo of a leopard, a compact cat noticeably smaller and lower slung than a lion”
	tiger	1. “a photo of a leopard, marked with clustered rosette spots rather than the long stripes of a tiger” 2. “a photo of a leopard, whose pale golden coat lacks the deep burnt orange tone of a tiger” 3. “a photo of a leopard, a smaller slender cat beside the massive build of a tiger”
	wolf	1. “a photo of a leopard, with a rounded feline face unlike the long canine muzzle of a wolf” 2. “a photo of a leopard, its spotted golden coat set apart from the plain gray fur of a wolf” 3. “a photo of a leopard, moving on soft padded paws with retractable claws no wolf has”

Table B.1 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
lion	bear	1. “a photo of a lion, the male framed by a thick shaggy mane that no bear grows” 2. “a photo of a lion, with a short sandy coat instead of the long shaggy fur of a bear” 3. “a photo of a lion, whose long tail ends in a dark tuft unlike the stub of a bear”
	leopard	1. “a photo of a lion, wearing a plain tawny coat free of the rosette spots of a leopard” 2. “a photo of a lion, its head circled by a heavy mane that a leopard never has” 3. “a photo of a lion, a far larger and more powerfully built cat than a leopard”
	tiger	1. “a photo of a lion, whose even sandy coat carries none of the black stripes of a tiger” 2. “a photo of a lion, with a dense mane framing the face unlike a bare headed tiger” 3. “a photo of a lion, showing a dark tuft at the tail tip that is absent on a tiger”
	wolf	1. “a photo of a lion, a large cat with a broad rounded face unlike the pointed snout of a wolf” 2. “a photo of a lion, whose mane and tawny coat look nothing like the gray fur of a wolf” 3. “a photo of a lion, far heavier and more muscular than a lean long legged wolf”
tiger	bear	1. “a photo of a tiger, striped in bold black bands that never appear on a bear” 2. “a photo of a tiger, with a burning orange coat instead of the plain brown fur of a bear” 3. “a photo of a tiger, a long low slung cat rather than a tall hunched bear”
	leopard	1. “a photo of a tiger, marked with long vertical stripes rather than the rosettes of a leopard” 2. “a photo of a tiger, whose deep orange coat is far richer than the pale gold of a leopard” 3. “a photo of a tiger, a massive heavy cat beside the slender frame of a leopard”
	lion	1. “a photo of a tiger, banded with black stripes that a plain coated lion entirely lacks” 2. “a photo of a tiger, with a bare maneless head unlike a male lion” 3. “a photo of a tiger, showing white markings on the face and belly that are absent on a lion”

Table B.1 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	wolf	1. “a photo of a tiger, a striped orange cat utterly unlike a gray furred wolf” 2. “a photo of a tiger, with a broad rounded feline face instead of the long muzzle of a wolf” 3. “a photo of a tiger, vastly heavier and more powerfully built than a slender wolf”
wolf	bear	1. “a photo of a wolf, a lean long legged animal far lighter than a bulky bear” 2. “a photo of a wolf, with sharp upright ears unlike the small rounded ears of a bear” 3. “a photo of a wolf, carrying a long bushy tail that a short tailed bear lacks”
	leopard	1. “a photo of a wolf, with a long dog like muzzle unlike the flat feline face of a leopard” 2. “a photo of a wolf, its grizzled gray coat free of any of the rosettes of a leopard” 3. “a photo of a wolf, running on blunt fixed claws that a leopard would keep sheathed”
	lion	1. “a photo of a wolf, a canine with a narrow pointed face unlike a broad headed lion” 2. “a photo of a wolf, whose gray coat and bushy tail replace the tawny mane of a lion” 3. “a photo of a wolf, lighter and leggier than a heavily muscled lion”
	tiger	1. “a photo of a wolf, its plain gray coat carrying none of the stripes of a tiger” 2. “a photo of a wolf, a dog like animal with a long snout unlike the short muzzle of a tiger” 3. “a photo of a wolf, far smaller and leaner than a heavyset tiger”

B.2 *mini*ImageNet

The *mini*ImageNet matrix covers five hound and gundog breeds from the training split: Walker hound, Ibizan hound, Saluki, Gordon setter, and golden retriever. It has shape $(5 \times 5 \times 3)$, with 20 directed pairs and 60 prompts in total (Table B.2).

Table B.2: Discriminative prompt matrix M for *mini*ImageNet. Each row lists the $k = 3$ prompts of $M_{i,j}$.

Anchor class i	Rival class j	Prompts in $M_{i,j}$
Walker hound	Ibizan hound	1. "a photo of a Walker hound, with long pendulous ears that hang down unlike the tall erect ears of an Ibizan hound" 2. "a photo of a Walker hound, its white coat broken by a black saddle and tan patches, not the red and white of an Ibizan hound" 3. "a photo of a Walker hound, a solidly built scent hound rather than the racy fine boned Ibizan hound"
	Saluki	1. "a photo of a Walker hound, with a short close lying coat lacking the silky feathering of a Saluki" 2. "a photo of a Walker hound, showing a broad blocky muzzle unlike the narrow tapered head of a Saluki" 3. "a photo of a Walker hound, tricolored in black white and tan rather than the pale cream of a Saluki"
	Gordon setter	1. "a photo of a Walker hound, whose short smooth coat has none of the long silky fringe of a Gordon setter" 2. "a photo of a Walker hound, marked with large white areas that are absent on a coal black Gordon setter" 3. "a photo of a Walker hound, a lighter framed scent hound beside the heavier boned Gordon setter"
	golden retriever	1. "a photo of a Walker hound, patched in black white and tan rather than the solid gold of a golden retriever" 2. "a photo of a Walker hound, with a short flat coat unlike the dense feathered fur of a golden retriever" 3. "a photo of a Walker hound, leaner and more angular than a stocky golden retriever"
Ibizan hound	Walker hound	1. "a photo of an Ibizan hound, with huge upright pointed ears that a drop eared Walker hound lacks" 2. "a photo of an Ibizan hound, colored red and white without the black saddle of a Walker hound" 3. "a photo of an Ibizan hound, tall and finely boned beside a sturdier Walker hound"
	Saluki	1. "a photo of an Ibizan hound, whose ears stand straight up unlike the feathered drop ears of a Saluki" 2. "a photo of an Ibizan hound, with a short hard coat instead of the silky fringe of a Saluki" 3. "a photo of an Ibizan hound, showing a pink nose and amber eyes unlike a dark nosed Saluki"

Table B.2 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	Gordon setter	1. “a photo of an Ibizan hound, with erect pointed ears unlike the long drop ears of a Gordon setter” 2. “a photo of an Ibizan hound, coated in short red and white rather than the black and tan of a Gordon setter” 3. “a photo of an Ibizan hound, lean and leggy beside a heavier boned Gordon setter”
	golden retriever	1. “a photo of an Ibizan hound, its tall erect ears nothing like the folded ears of a golden retriever” 2. “a photo of an Ibizan hound, with a thin short coat unlike the thick golden fur of a golden retriever” 3. “a photo of an Ibizan hound, narrow and deep chested rather than broad bodied like a golden retriever”
Saluki	Walker hound	1. “a photo of a Saluki, with long silky feathering on the ears and tail that is absent on a Walker hound” 2. “a photo of a Saluki, a slender deep chested sighthound unlike a stockier Walker hound” 3. “a photo of a Saluki, pale cream or fawn rather than the black white and tan of a Walker hound”
	Ibizan hound	1. “a photo of a Saluki, whose long feathered ears hang down unlike the erect ears of an Ibizan hound” 2. “a photo of a Saluki, with silky fringed fur instead of the short hard coat of an Ibizan hound” 3. “a photo of a Saluki, showing a dark nose and dark eyes unlike a pink nosed Ibizan hound”
	Gordon setter	1. “a photo of a Saluki, an extremely slender sighthound beside a solid muscular Gordon setter” 2. “a photo of a Saluki, pale cream or fawn rather than the glossy black and tan of a Gordon setter” 3. “a photo of a Saluki, with a deeply tucked waist and narrow head unlike a Gordon setter”
	golden retriever	1. “a photo of a Saluki, whose narrow waist and deep chest contrast the blocky body of a golden retriever” 2. “a photo of a Saluki, with sparse silky feathering unlike the thick water repellent coat of a golden retriever” 3. “a photo of a Saluki, having a long tapered skull rather than the broad head of a golden retriever”
Gordon setter	Walker hound	1. “a photo of a Gordon setter, with a long silky feathered coat unlike the short fur of a Walker hound” 2. “a photo of a Gordon setter, coal black with tan points and none of the white patches of a Walker hound” 3. “a photo of a Gordon setter, a heavier built gundog beside a lighter framed Walker hound”

Table B.2 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	Ibizan hound	1. “a photo of a Gordon setter, with long drop ears unlike the upright pointed ears of an Ibizan hound” 2. “a photo of a Gordon setter, its glossy black and tan coat unlike the red and white of an Ibizan hound” 3. “a photo of a Gordon setter, solidly muscled rather than lean and racy like an Ibizan hound”
	Saluki	1. “a photo of a Gordon setter, deep black with rich tan markings unlike a pale cream Saluki” 2. “a photo of a Gordon setter, a sturdy muscular gundog beside a fragile looking Saluki” 3. “a photo of a Gordon setter, with a broad squared muzzle unlike the narrow head of a Saluki”
	golden retriever	1. “a photo of a Gordon setter, black with rich tan markings rather than the uniform gold of a golden retriever” 2. “a photo of a Gordon setter, whose longer leaner frame contrasts the stocky build of a golden retriever” 3. “a photo of a Gordon setter, with a narrower head than a broad skulled golden retriever”
golden retriever	Walker hound	1. “a photo of a golden retriever, in a solid golden coat without the tricolor patches of a Walker hound” 2. “a photo of a golden retriever, with thick feathered fur unlike the short smooth coat of a Walker hound” 3. “a photo of a golden retriever, broad and sturdy beside a leaner Walker hound”
	Ibizan hound	1. “a photo of a golden retriever, whose ears fold close to the head unlike the erect ears of an Ibizan hound” 2. “a photo of a golden retriever, densely coated in gold rather than thin short red and white” 3. “a photo of a golden retriever, heavy and broad chested unlike a slim racy Ibizan hound”
	Saluki	1. “a photo of a golden retriever, with a broad blocky head unlike the narrow skull of a Saluki” 2. “a photo of a golden retriever, thickly coated in gold rather than the sparse silky fur of a Saluki” 3. “a photo of a golden retriever, solidly built without the deeply tucked waist of a Saluki”
	Gordon setter	1. “a photo of a golden retriever, uniformly gold rather than black with tan like a Gordon setter” 2. “a photo of a golden retriever, with a broader heavier head than a Gordon setter” 3. “a photo of a golden retriever, stockier and shorter coupled than a longer framed Gordon setter”

B.3 CUB-200-2011

The CUB-200-2011 matrix covers five gull species (CUB classes 059–064): California Gull, Glaucous-winged Gull, Heermann's Gull, Herring Gull, and Ring-billed Gull. It has shape $(5 \times 5 \times 3)$, with 20 directed pairs and 60 prompts in total (Table B.3).

Table B.3: Discriminative prompt matrix M for CUB-200-2011. Each row lists the $k = 3$ prompts of $M_{i,j}$.

Anchor class i	Rival class j	Prompts in $M_{i,j}$
California Gull	Glaucous-winged Gull	1. "a photo of a California Gull, with black wingtips that a pale winged Glaucous-winged Gull never shows" 2. "a photo of a California Gull, standing on greenish yellow legs rather than the pink legs of a Glaucous-winged Gull" 3. "a photo of a California Gull, a medium sized gull much smaller than a bulky Glaucous-winged Gull"
	Heermann's Gull	1. "a photo of a California Gull, with white underparts unlike the sooty gray body of a Heermann's Gull" 2. "a photo of a California Gull, its yellow bill unlike the bright coral red bill of a Heermann's Gull" 3. "a photo of a California Gull, showing greenish legs rather than the black legs of a Heermann's Gull"
	Herring Gull	1. "a photo of a California Gull, with greenish yellow legs instead of the pink legs of a Herring Gull" 2. "a photo of a California Gull, having a dark eye unlike the pale staring eye of a Herring Gull" 3. "a photo of a California Gull, whose bill carries both a black mark and a red spot unlike a Herring Gull"
	Ring-billed Gull	1. "a photo of a California Gull, larger with a darker gray mantle than a Ring-billed Gull" 2. "a photo of a California Gull, its bill marked with red as well as black rather than a clean black ring" 3. "a photo of a California Gull, with a dark eye rather than the pale eye of a Ring-billed Gull"
Glaucous-winged Gull	California Gull	1. "a photo of a Glaucous-winged Gull, whose wingtips are pale gray without any of the black of a California Gull" 2. "a photo of a Glaucous-winged Gull, standing on pink legs rather than the greenish legs of a California Gull" 3. "a photo of a Glaucous-winged Gull, a heavy bulky gull far larger than a California Gull"
	Heermann's Gull	1. "a photo of a Glaucous-winged Gull, pale gray and white unlike the dark sooty Heermann's Gull" 2. "a photo of a Glaucous-winged Gull, with a yellow bill instead of the red black tipped bill of a Heermann's Gull" 3. "a photo of a Glaucous-winged Gull, on pink legs rather than the black legs of a Heermann's Gull"

Table B.3 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	Herring Gull	1. "a photo of a Glaucous-winged Gull, whose wingtips match the mantle instead of the black tips of a Herring Gull" 2. "a photo of a Glaucous-winged Gull, with a dark eye unlike the pale yellow eye of a Herring Gull" 3. "a photo of a Glaucous-winged Gull, bulkier with a much heavier bill than a Herring Gull"
	Ring-billed Gull	1. "a photo of a Glaucous-winged Gull, a very large gull dwarfing a small Ring-billed Gull" 2. "a photo of a Glaucous-winged Gull, with pale gray wingtips unlike the black tips of a Ring-billed Gull" 3. "a photo of a Glaucous-winged Gull, on pink legs rather than the bright yellow legs of a Ring-billed Gull"
Heermann's Gull	California Gull	1. "a photo of a Heermann's Gull, its body dark sooty gray rather than the white of a California Gull" 2. "a photo of a Heermann's Gull, with a bright coral red bill unlike a yellow billed California Gull" 3. "a photo of a Heermann's Gull, standing on black legs rather than greenish yellow ones"
	Glaucous-winged Gull	1. "a photo of a Heermann's Gull, uniformly dark gray unlike a very pale Glaucous-winged Gull" 2. "a photo of a Heermann's Gull, with a red black tipped bill rather than a plain yellow bill" 3. "a photo of a Heermann's Gull, a slim medium sized gull beside a massive Glaucous-winged Gull"
	Herring Gull	1. "a photo of a Heermann's Gull, with sooty gray underparts where a Herring Gull is clean white" 2. "a photo of a Heermann's Gull, its coral red bill unlike the yellow bill of a Herring Gull" 3. "a photo of a Heermann's Gull, on black legs instead of the pink legs of a Herring Gull"
	Ring-billed Gull	1. "a photo of a Heermann's Gull, dark gray bodied unlike a white breasted Ring-billed Gull" 2. "a photo of a Heermann's Gull, with a red bill tipped in black rather than a black ringed yellow bill" 3. "a photo of a Heermann's Gull, standing on black legs rather than bright yellow legs"
Herring Gull	California Gull	1. "a photo of a Herring Gull, standing on bright pink legs unlike a greenish legged California Gull" 2. "a photo of a Herring Gull, with a pale staring yellow eye rather than the dark eye of a California Gull" 3. "a photo of a Herring Gull, a large heavy gull noticeably bigger than a California Gull"

Table B.3 continued

Anchor class i	Rival class j	Prompts in $M_{i,j}$
	Glaucous-winged Gull	1. "a photo of a Herring Gull, with black wingtips marked by white spots unlike pale gray wingtips" 2. "a photo of a Herring Gull, its pale yellow eye unlike the dark eye of a Glaucous-winged Gull" 3. "a photo of a Herring Gull, slimmer with a lighter bill than a Glaucous-winged Gull"
	Heermann's Gull	1. "a photo of a Herring Gull, clean white below unlike the sooty gray body of a Heermann's Gull" 2. "a photo of a Herring Gull, with a yellow bill bearing a red spot rather than a coral red bill" 3. "a photo of a Herring Gull, on pink legs instead of the black legs of a Heermann's Gull"
	Ring-billed Gull	1. "a photo of a Herring Gull, much larger and heavier billed than a Ring-billed Gull" 2. "a photo of a Herring Gull, with pink legs rather than the bright yellow legs of a Ring-billed Gull" 3. "a photo of a Herring Gull, its bill marked by a red spot and not by a black ring"
Ring-billed Gull	California Gull	1. "a photo of a Ring-billed Gull, with a crisp black ring around the bill and no red spot" 2. "a photo of a Ring-billed Gull, standing on bright yellow legs unlike the greenish legs of a California Gull" 3. "a photo of a Ring-billed Gull, smaller with a paler gray mantle than a California Gull"
	Glaucous-winged Gull	1. "a photo of a Ring-billed Gull, a small slight gull beside a huge Glaucous-winged Gull" 2. "a photo of a Ring-billed Gull, with black wingtips unlike the pale gray wingtips of a Glaucous-winged Gull" 3. "a photo of a Ring-billed Gull, on bright yellow legs rather than pink legs"
	Heermann's Gull	1. "a photo of a Ring-billed Gull, white headed and white breasted unlike a dark sooty Heermann's Gull" 2. "a photo of a Ring-billed Gull, with a yellow bill ringed in black rather than a coral red bill" 3. "a photo of a Ring-billed Gull, standing on yellow legs instead of black legs"
	Herring Gull	1. "a photo of a Ring-billed Gull, notably smaller and more delicate than a Herring Gull" 2. "a photo of a Ring-billed Gull, with yellow legs rather than the pink legs of a Herring Gull" 3. "a photo of a Ring-billed Gull, its bill circled by a black ring without any red spot"